

SATELLITES, NAVIC, GAGAN AND APPLICATIONS

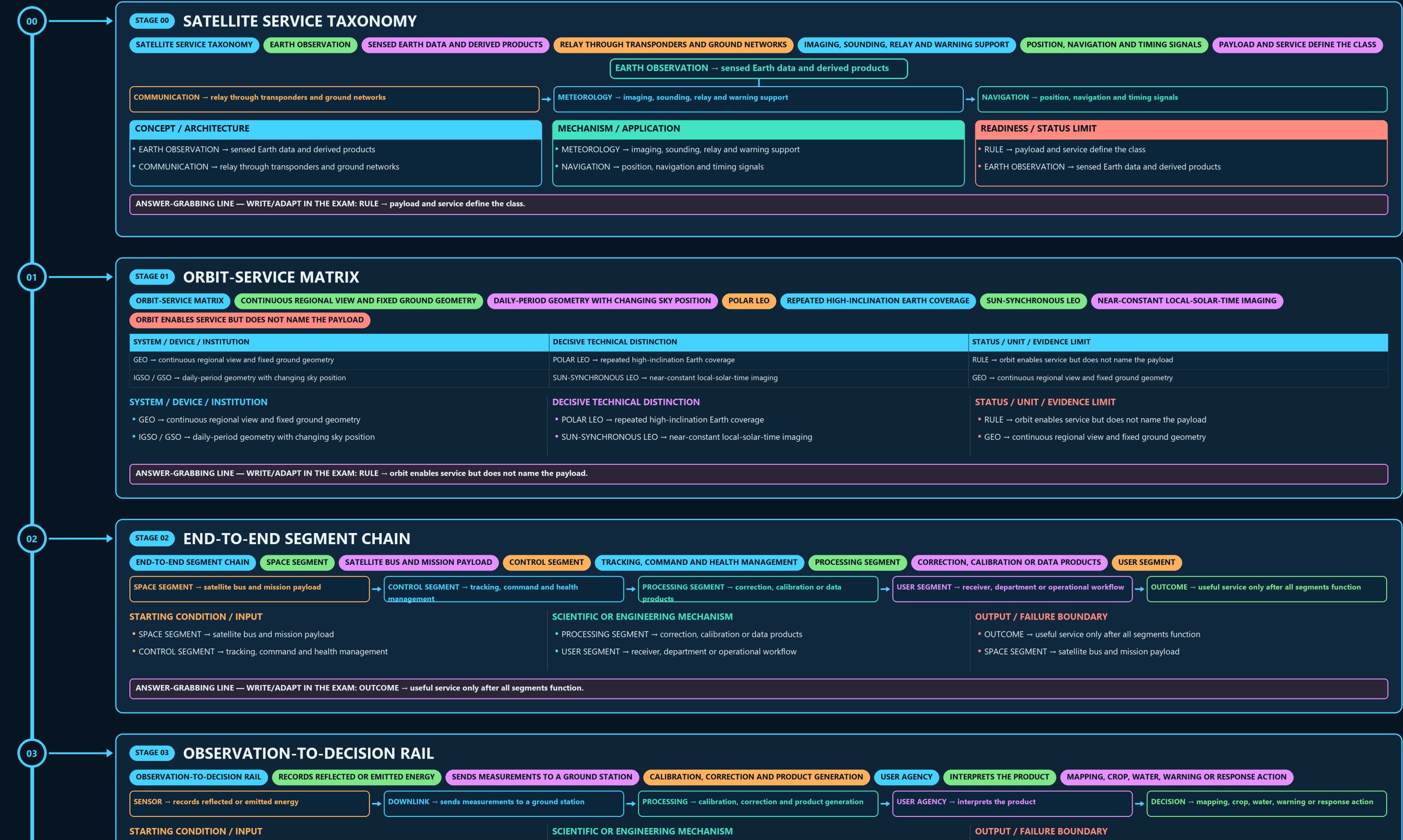
SATELLITE SERVICE TAXONOMY → ORBIT-SERVICE MATRIX → END-TO-END SEGMENT CHAIN → OBSERVATION-TO-DECISION RAIL → COMMUNICATION RELAY LOOP → NAVIC SYSTEM MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



06

STAGE 06 NAVIC SERVICE AND SIGNAL STACK

NAVIC SERVICE AND SIGNAL STACK

OPEN POSITIONING SERVICE

ENCRYPTED SERVICE FOR AUTHORISED USERS

ORIGINAL SIGNAL LAYERS IN OWNER

MASS-MARKET INTEROPERABILITY DIRECTION

PAYLOAD SUPPORT DOES NOT PROVE DEVICE ADOPTION

SPS → open positioning service

RS → encrypted service for authorised users

LS / S → original signal layers in owner

L1 → mass-market interoperability direction

CONCEPT / ARCHITECTURE

- SPS → open positioning service
- RS → encrypted service for authorised users

MECHANISM / APPLICATION

- LS / S → original signal layers in owner
- L1 → mass-market interoperability direction

READINESS / STATUS LIMIT

- RULE → payload support does not prove device adoption
- SPS → open positioning service

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → payload support does not prove device adoption.

07

STAGE 07 CONSTELLATION-HEALTH FIREWALL

CONSTELLATION-HEALTH FIREWALL

CUMULATIVE MISSION COUNT

INTENDED ORBIT

ORBIT-RAISING COMPLETED

SPACECRAFT PERFORMING A STATED FUNCTION

PNT CAPABLE

FULL NAVIGATION SERVICE CONTRIBUTION

MESSAGE BROADCAST

SYSTEM / DEVICE / INSTITUTION

LAUNCHED → cumulative mission count

INTENDED ORBIT → orbit-raising completed

DECISIVE TECHNICAL DISTINCTION

OPERATIONAL → spacecraft performing a stated function

PNT CAPABLE → full navigation service contribution

STATUS / UNIT / EVIDENCE LIMIT

MESSAGE BROADCAST → useful but not equivalent to PNT

LAUNCHED → cumulative mission count

SYSTEM / DEVICE / INSTITUTION

- LAUNCHED → cumulative mission count
- INTENDED ORBIT → orbit-raising completed

DECISIVE TECHNICAL DISTINCTION

- OPERATIONAL → spacecraft performing a stated function
- PNT CAPABLE → full navigation service contribution

STATUS / UNIT / EVIDENCE LIMIT

- MESSAGE BROADCAST → useful but not equivalent to PNT
- LAUNCHED → cumulative mission count

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MESSAGE BROADCAST → useful but not equivalent to PNT.

08

STAGE 08 GAGAN CORRECTION CHAIN

GAGAN CORRECTION CHAIN

REFERENCE STATIONS

MEASURE GNSS ERROR

MASTER CONTROL

COMPUTES CORRECTION AND INTEGRITY

UPLINK STATIONS

SEND AUGMENTATION MESSAGE

GEO PAYLOAD

REFERENCE STATIONS → measure GNSS error

MASTER CONTROL → computes correction and integrity

UPLINK STATIONS → send augmentation message

GEO PAYLOAD → rebroadcasts over a wide area

AIRCRAFT RECEIVER → applies correction and checks integrity

STARTING CONDITION / INPUT

- REFERENCE STATIONS → measure GNSS error
- MASTER CONTROL → computes correction and integrity

SCIENTIFIC OR ENGINEERING MECHANISM

- UPLINK STATIONS → send augmentation message
- GEO PAYLOAD → rebroadcasts over a wide area

OUTPUT / FAILURE BOUNDARY

- AIRCRAFT RECEIVER → applies correction and checks integrity
- REFERENCE STATIONS → measure GNSS error

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AIRCRAFT RECEIVER → applies correction and checks integrity.

09

STAGE 09 ACCURACY-INTEGRITY GATE

ACCURACY-INTEGRITY GATE

ESTIMATED POSITION CLOSE TO TRUE POSITION

WARNING WHEN INFORMATION MUST NOT BE USED

TIME TO ALERT

WARNING MUST ARRIVE WITHIN A SAFE INTERVAL

AVIATION USE

CERTIFICATION GOVERNS OPERATIONAL ACCEPTANCE

A PRECISE SIGNAL WITHOUT INTEGRITY IS UNSAFE

SYSTEM / DEVICE / INSTITUTION

ACCURACY → estimated position close to true position

INTEGRITY → warning when information must not be used

DECISIVE TECHNICAL DISTINCTION

TIME TO ALERT → warning must arrive within a safe interval

AVIATION USE → certification governs operational acceptance

STATUS / UNIT / EVIDENCE LIMIT

RULE → a precise signal without integrity is unsafe

MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

SYSTEM / DEVICE / INSTITUTION

- ACCURACY → estimated position close to true position
- INTEGRITY → warning when information must not be used

DECISIVE TECHNICAL DISTINCTION

- TIME TO ALERT → warning must arrive within a safe interval
- AVIATION USE → certification governs operational acceptance

STATUS / UNIT / EVIDENCE LIMIT

- RULE → a precise signal without integrity is unsafe
- MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a precise signal without integrity is unsafe.

A PRECISE SIGNAL WITHOUT INTEGRITY IS UNSAFE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
ACCURACY → estimated position close to true position	TIME TO ALERT → warning must arrive within a safe interval	RULE → a precise signal without integrity is unsafe
INTEGRITY → warning when information must not be used	AVIATION USE → certification governs operational acceptance	MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

SYSTEM / DEVICE / INSTITUTION

• ACCURACY → estimated position close to true position

• INTEGRITY → warning when information must not be used

DECISIVE TECHNICAL DISTINCTION

• TIME TO ALERT → warning must arrive within a safe interval

• AVIATION USE → certification governs operational acceptance

STATUS / UNIT / EVIDENCE LIMIT

• RULE → a precise signal without integrity is unsafe

• MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a precise signal without integrity is unsafe.

10

STAGE 10

SBAS-GBAS INSTITUTION MAP

SBAS-GBAS INSTITUTION MAP

WIDE-AREA CORRECTIONS REBROADCAST FROM SATELLITES

LOCAL AIRPORT CORRECTIONS BROADCAST FROM THE GROUND

ISRO + AAI

GAGAN DEVELOPMENT PARTNERSHIP

AVIATION OPERATIONAL ROLE

AVIATION CERTIFICATION ROLE

CLOSE DISTINCTION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SBAS → wide-area corrections rebroadcast from satellites	ISRO + AAI → GAGAN development partnership	DGCA → aviation certification role
GBAS → local airport corrections broadcast from the ground	AAI → aviation operational role	CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is...

SYSTEM / DEVICE / INSTITUTION

• SBAS → wide-area corrections rebroadcast from satellites

• GBAS → local airport corrections broadcast from the ground

DECISIVE TECHNICAL DISTINCTION

• ISRO + AAI → GAGAN development partnership

• AAI → aviation operational role

STATUS / UNIT / EVIDENCE LIMIT

• DGCA → aviation certification role

• CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is.

11

STAGE 11

PUBLIC-VALUE ANSWER SPINE

PUBLIC-VALUE ANSWER SPINE

OBSERVATION COMMUNICATION METEOROLOGY PNT OR AUGMENTATION

SPACE GROUND PROCESSING AND USER SEGMENTS

NAVIC CONSTELLATION FROM GAGAN SBAS

STANDARDS ADOPTION INTEGRITY PRIVACY JAMMING AND SPOOFING

DATED HEALTH FREQUENCY COVERAGE ACCURACY AND CERTIFICATION

EVIDENCE LIMIT

STATE CONSTELLATION/SERVICE AREA, ORBIT

CLASSIFY → observation communication meteorology PNT or augmentation

TRACE → space ground processing and user segments

DISTINGUISH → NavIC constellation from GAGAN SBAS

TEST → standards adoption integrity privacy jamming and spoofing

VERIFY → dated health frequency coverage accuracy and certification

CONCEPT / ARCHITECTURE

• CLASSIFY → observation communication meteorology PNT or augmentation

• TRACE → space ground processing and user segments

MECHANISM / APPLICATION

• DISTINGUISH → NavIC constellation from GAGAN SBAS

• TEST → standards adoption integrity privacy jamming and spoofing

READINESS / STATUS LIMIT

• VERIFY → dated health frequency coverage accuracy and certification

• EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

DEVELOPS AND OPERATES SATELLITE PLATFORMS, THE NAVIC SPACE AND

ISRO BUILDS AND OPERATES THE NAVIGATION CONSTELLATION — IT

AAI + ISRO

JOINT DEVELOPERS OF GAGAN.

AAI IS THE OPERATIONAL OWNER ON THE AVIATION SIDE

USER MINISTRIES/AGENCIES SUCH AS IMD AND OTHER MOES BODIES

CONVERT METEOROLOGICAL SATELLITE INPUTS INTO ACTIONABLE FORECASTS AND WARNINGS.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

• ISRO: develops and operates satellite platforms, the NavIC space and control segment and major application systems.

• (Precisely: ISRO builds and operates the navigation constellation — it is not the sole "owner" of every downstream service built on it.)

• AAI + ISRO: joint developers of GAGAN.

• AAI is the operational owner on the aviation side, DGCA is the certifying regulator (RNP 0.1 in 2013,

READINESS, UNIT OR STATUS LIMIT

• User ministries/agencies such as IMD and other MoES bodies: convert meteorological satellite inputs into actionable forecasts and warnings.

• NRSC/Bhuvan ecosystem: exemplifies how data becomes policy utility.

• CAUTION: Navigation systems sit at the overlap of civilian convenience, critical infrastructure and strategic autonomy

• this makes standards, signal policy and device ecosystem choices important policy questions.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

• CAUTION: GAGAN's aviation utility shows that "space" is also an air-navigation governance issue, not only a scientific one.

• CAUTION: Broader NavIC adoption depends on procurement standards, handset/chipset support, automotive integration and inter-operability politics.

• CAUTION: An advanced answer can note that data accessibility, security restrictions and

• commercial receiver availability often determine actual impact more than satellite launch success.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.